



SOW Document to **Peer Competitor Analysis Dashboard**

Document Summary

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Project Name: Peer Competitor Analysis Dashboard

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1. Technical Evaluation

- **Received By:** Udbhav Mishra
- **Revision (V2.0):** 07/09/2026 - Architecture, Redbus marketplace intelligence, SRP snapshot model, and delivery scope
- **Date Assigned:** Based on sign-off

2. Project Overview

The objective of this initiative is to build a centralized **Peer Competitive Analysis Dashboard** that enables FreshBus to continuously monitor, benchmark, and analyze competitor performance across key customer-facing platforms and marketplace channels.

The dashboard consolidates publicly available customer feedback and marketplace observations from **Google Play Store, Apple App Store, Google Search Reviews, Redbus, and Abhibus** into a single analytical product-eliminating the need for manual platform-wise analysis.

This solution supports data-driven decision making by enabling stakeholders to:

- Monitor competitor ratings, review volumes, and customer sentiment from a single platform.
- Benchmark FreshBus against key competitors across digital touchpoints.
- Understand marketplace scale, competition intensity, and FreshBus search visibility (SRP) on Redbus and Abhibus.
- Drill from marketplace → route → operator → service → snapshot for investigation.
- Identify competitor strengths, weaknesses, and emerging customer pain points.
- Track historical performance trends and detect significant changes over time.
- Uncover opportunities for product, service, and operational improvements.
- Reduce manual effort in collecting and analyzing competitive intelligence.
- Establish a scalable foundation for AI-generated business insights and conversational analytics.

The dashboard serves as the centralized source of competitive intelligence for strategic planning, customer experience improvements, and market positioning.

3. Problem Statement

Competitive performance data is distributed across multiple public platforms, making it difficult to monitor, compare, and analyze competitors efficiently.

Key challenges include:

- No centralized view of competitor ratings, reviews, and customer sentiment.
- Manual effort required to collect and compare data across platforms.

- Limited visibility into competitor trends and performance over time.
- No standardized benchmarking of FreshBus against competitors.
- Absence of route-level competitive intelligence and consolidated customer feedback analysis.
- No structured view of marketplace competition (operator density, service counts, fare position).
- No reliable historical record of Search Rank Position (SRP) by scheduled snapshot.
- Delayed identification of market opportunities and competitive risks.

As a result, competitive analysis is time-consuming, fragmented, and less effective for supporting timely business decisions.

4. Scope

The Peer Competitive Analysis Dashboard provides a centralized platform to analyze and benchmark competitor performance across customer-facing platforms and marketplace channels.

4.1 Overview Dashboard

- Unified competitive summary across all tracked sources.
- Cross-platform benchmarking and operator leaderboards.
- Historical trends and coverage indicators.
- Navigation into each analytical channel (Play, iOS, Google Reviews, Redbus, Abhibus).

4.2 Android Reviews (Google Play Store)

- Overall ratings, review volumes, downloads / ratings counts.
- Customer sentiment and topic / KPI classification.
- Trend analysis and peer comparison.

4.3 iOS Reviews (Apple App Store)

- Overall ratings, ratings counts, and review analysis where available.
- Customer sentiment and topic / KPI classification.
- Trend analysis and peer comparison.

4.4 Google Search Reviews

- Google Business ratings, review analysis, and reputation tracking.
- Customer sentiment and peer comparison.

4.5 Redbus Analytics - Marketplace Intelligence

Redbus is delivered as a purpose-built marketplace intelligence product (not SRP-only). Primary navigation has exactly four sections. The default landing page is **Marketplace**.

4.5.1 Marketplace

- Six KPI pulse cards: Active Services, Operators, Routes, Avg SRP Rank, Top-10 Services, Avg Fare.
- Marketplace Trend - one chart with a metric selector (Average SRP Rank / Average Fare / Active Services / Active Operators).
- Route Competition heatmap - active operators by route × snapshot slot (05:00 / 11:00 / 17:00 / 23:00 IST).
- What Changed - deterministic insight list (maximum five items).
- Alerts / Opportunities - only when qualifying conditions exist.
- Route Leaderboard - bridge from overview to route investigation.

4.5.2 Routes (Route Intelligence)

- Route KPI row: Routes, Avg Operators / Route, Avg Services / Route, FreshBus Route Coverage.
- Route Leaderboard with sorting and route selection.
- Route Competitive Matrix (operators vs FreshBus average SRP; bubble size = FreshBus services).
- Route Detail: operator competitive table and FreshBus SRP band distribution.

4.5.3 Operators (Operator Intelligence)

- Operator Leaderboard (FreshBus highlighted / pinned).
- Operator Competitive Positioning scatter (median fare vs average SRP).
- Operator Detail with route performance table.

4.5.4 SRP Tracker

- Summary KPIs: Avg SRP Rank, Top-10, Top-30, 100+ services.
- SRP Rank Movement chart (inverted Y-axis; #1 at top).
- SRP Rank Distribution (Top 10 / 11-30 / 31-100 / 100+).
- SRP Rank Stability table (best / worst / volatility / Top-10 days).
- Intraday SRP Movement (05:00 / 11:00 / 17:00 / 23:00 columns).
- Detailed SRP Workbook - hierarchical Route → Operator → Service with expandable Month → Weeks → Dates → Snapshot slots.
- Service Detail drawer for focused investigation (SRP history + latest competitive position).

Redbus also captures route-wise review analysis, sentiment classification, and operational attributes including Average Rating, Number of Ratings, Bus Type, Bus Service Timing, Travel Duration, and SRP for FreshBus and eligible competitors. Route-level competitor inclusion follows the business rule of operators with three or more services on a route where that rule applies.

4.6 Abhibus Analytics

- Same information architecture as Redbus: Marketplace, Routes, Operators, SRP Tracker.
- Shared filter model, metric definitions, and snapshot slots adapted to Abhibus marketplace data.

- Scraping timeline matches Redbus: **5:00 AM, 11:00 AM, 5:00 PM, and 11:00 PM IST** (four snapshots per day).

4.7 Platform Capabilities

- Global header with data status and Sync control.
- Shared filters: Route, Operator, Period (Today / Yesterday / Last 7 days / Month to date / Custom), Compare, Snapshot.
- Data quality indicators (expected vs completed snapshots, partial coverage).
- Staging and Production environments; Git-based version control and controlled releases.
- AI Insights Engine and AI Conversational Assistant as delivery modules of this SOW.

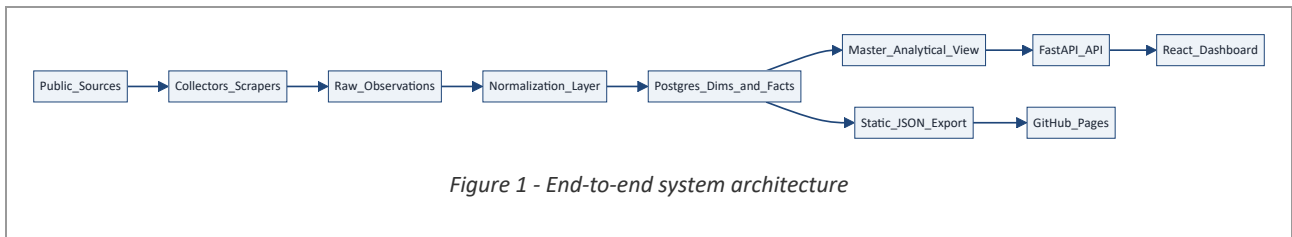
4.8 Out of Scope (this SOW)

The following are explicitly excluded from this SOW to preserve analytical transparency: unexplained “competition / health / visibility” scores without a documented formula; predictive rank forecasting; automated pricing or elasticity recommendations; sentiment analysis beyond the defined review-scoring model where not already specified in delivery modules.

5. Technical Approach

The Peer Competitive Analysis Dashboard is a web-based internal analytics application with an automated data collection, normalization, and analytical presentation pipeline.

5.1 System Architecture



Primary components:

- **scraper/** - Channel collectors (Playwright / Selenium), HTML capture, parsing, DAL helpers.
- **aggregator/** - Refresh orchestration, sentiment scoring, metrics calculation, validation.
- **api/** - FastAPI application exposing analytical and operational endpoints.
- **db/** - PostgreSQL schema, migrations, seed operators and routes.
- **dashboard/** - React + Vite frontend (Recharts, shared filter context).
- **docker/** - Container definitions for API, scraper, dashboard, and Postgres.

5.2 Data Collection

Publicly available data is collected using automated web scraping implemented in Python.

Channel	Method	Cadence (IST)	Primary outputs
Google Play Store	App store collector (Python)	Daily 08:00	Rating, reviews, downloads, topic scores
Apple App Store	App store collector (Python)	Daily 08:00	Rating, ratings count, reviews where available
Google Search Reviews	Automated browser collector	Daily 08:00	Business rating, reviews, sentiment inputs
Redbus - Reviews	Playwright (Chromium)	Per refresh cycle	Route/operator ratings and review text
Redbus - SRP / Listings	Selenium → HTML save → parse	5:00 AM, 11:00 AM, 5:00 PM, 11:00 PM IST	SRP rank, fare, bus type, timing, duration, rating
Abhibus - SRP / Listings	Parallel marketplace collectors (same snapshot model as Redbus)	5:00 AM, 11:00 AM, 5:00 PM, 11:00 PM IST	Same analytical grain as Redbus marketplace

5.3 Redbus SRP / Marketplace Scraping Flow

Marketplace intelligence is **snapshot-based** (not live). Each collection window is identified by **snapshot date + scheduled snapshot slot (5:00 AM / 11:00 AM / 5:00 PM / 11:00 PM IST)**. This schedule applies to both Redbus and Abhibus. Actual scrape completion time is stored separately and is not used as the analytical time dimension.

- 1. Configure routes** - Origin-destination pairs are loaded from route configuration (e.g. Vijayawada → Visakhapatnam and other tracked corridors).
- 2. Open Redbus search results** - Selenium Chrome navigates to the bus-ticket results page for the travel date and route.
- 3. Scroll to end of list** - The scraper scrolls until the marketplace list is fully loaded ("End of list"), so ranks reflect full result order.
- 4. Save raw HTML** - Page HTML is retained under scraper/redbus_html/ for auditability and re-parse without re-hitting the site.
- 5. Parse service cards** - Extract operator name, boarding/dropping times, rating, review count, final fare, bus type, duration, and **SRP rank** (position in results).
- 6. Persist immutable observations** - One observation per service × snapshot. Historical rows are never overwritten by a later scrape of a different slot.

7. **Normalize into analytical layer** - Map into routes, operators, services, snapshots, and service_snapshot_facts; expose via master analytical view to the dashboard.

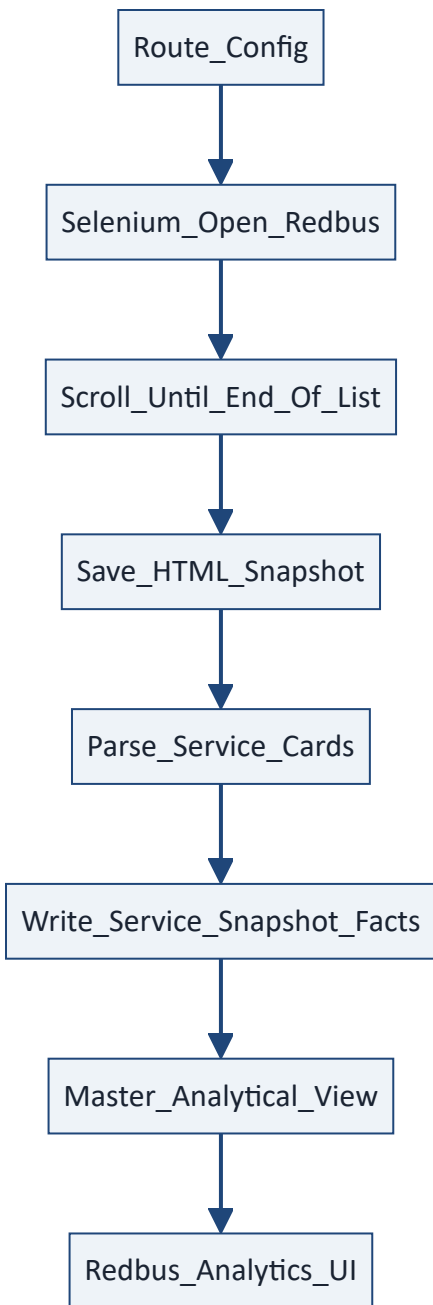


Figure 2 - Redbus SRP / marketplace scrape pipeline

Fields collected for marketplace observations

Category	Fields
Snapshot	Snapshot date, scheduled slot, scraped_at, snapshot status (SUCCESS / PARTIAL / FAILED)
Route	Origin, destination, route name / key
Operator	Operator identity and display name
Service	Service identity, service name, bus type, departure, arrival, travel duration
SRP	srp_rank, total results (where available), service_present flag
Pricing	Base fare, discount, final / displayed fare
Availability	Total seats, available seats (when reliably present on page)
Ratings	Rating, rating count

Important: The scraper captures observations only. Business metrics (averages, rank change, fare gap, Top-10 counts) are computed in the analytical layer-not inside the scraper.

5.4 Snapshot Schedule (Marketplace) - Redbus & Abhibus

Both **Redbus** and **Abhibus** marketplace / SRP data are collected on the same daily timeline. Four expected snapshots per day:

5:00 AM IST Morning snapshot Redbus + Abhibus	11:00 AM IST Midday snapshot Redbus + Abhibus	5:00 PM IST Evening snapshot Redbus + Abhibus	11:00 PM IST Night snapshot Redbus + Abhibus
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Slot (IST)	24-hour label	Channels
5:00 AM	05:00	Redbus marketplace / SRP · Abhibus marketplace / SRP
11:00 AM	11:00	Redbus marketplace / SRP · Abhibus marketplace / SRP
5:00 PM	17:00	Redbus marketplace / SRP · Abhibus marketplace / SRP
11:00 PM	23:00	Redbus marketplace / SRP · Abhibus marketplace / SRP

Incomplete or failed snapshots are visible in data-quality status and are excluded from averages when status is FAILED. Partial snapshots are labelled, not silently treated as complete coverage.

5.5 Sync vs Scrape

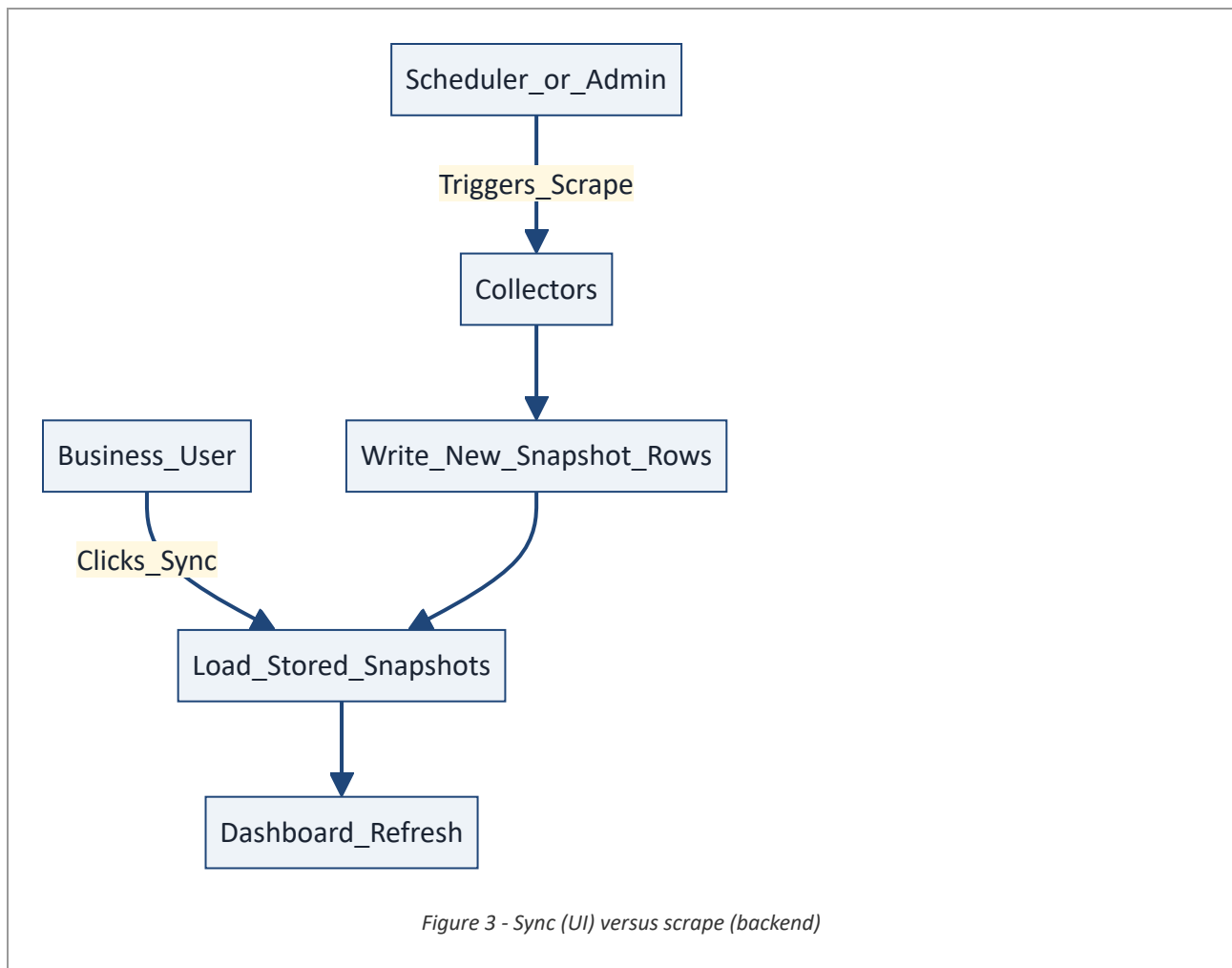
Sync (Dashboard button)

- Reloads / refreshes **stored** analytical snapshots.
- **Does not** start a Redbus or marketplace scrape.

Scrape (Backend only)

- Triggered by scheduler or authorized admin refresh API.

- Tooltip: “Sync loads stored snapshots - never triggers scrape.”
- Produces new immutable snapshot observations.
- Never initiated from the business user’s Sync control.



5.6 Data Model - Analytical Grain

Recommended architecture: Raw scraped data → Normalization → Dimension tables + Fact table → Master analytical view → Dashboard.

Entity	Role
routes	Origin-destination corridor identity
operators	Operator identity (FreshBus and peers)
services	Stable service attributes on a route
snapshots	Collection window (date + slot + status)
service_snapshot_facts	One row = one service × one snapshot

Mandatory null rule: Missing SRP is not converted to 999. Missing fare is not converted to 0. Missing values are excluded from averages. Absence of a service is tracked via `service_present`, not fake ranks.

5.7 Dashboard Application

- Browser-based SPA using **React + Vite**, charts via Recharts.
- Desktop-first analytical UI (target 1440px+), shared global filters across Redbus / Abhibus pages.
- Marketplace tracking start date: **01 September 2026** (hard business boundary for date pickers).
- All business-facing timestamps displayed in **IST** (24-hour operational times for scrape slots).
- API served on port **8000**; dashboard on port **3000** in local / container deployments.
- Static export path supports GitHub Pages hosting of the frontend with pre-built analytical JSON where required.

5.8 Information Architecture - User Journey

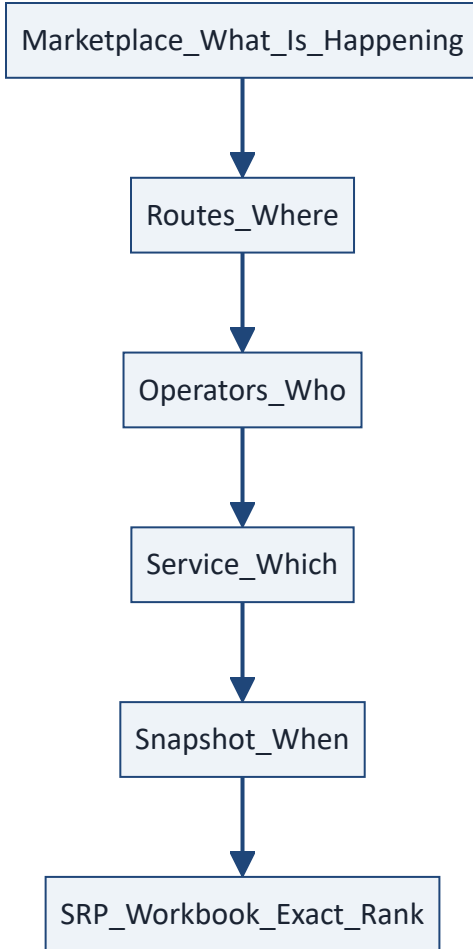


Figure 4 - Marketplace intelligence drill-down journey

5.9 Core Metric Definitions

Metric	Definition
Active Routes	Count of distinct routes with at least one observed service in the selected snapshot / period.
Active Operators	Count of distinct operators with at least one observed service. Operators with no observation are not active.
Active Services	Count of distinct services observed in the selected snapshot / period.
Average SRP Rank	Arithmetic mean of observed SRP ranks. Lower number = better. NULLs excluded. Display example: #24.8.
Top-10 / Top-30 Services	Services whose average SRP rank for the period is ≤ 10 / ≤ 30 .
SRP Rank Change	Previous average – Current average. Positive change = improvement. Display: “↑ 6 positions better”.
Average / Median Fare	Mean or median of observed displayed / final fares (valid fares only).
Fare Change %	$(\text{Current avg} - \text{Previous avg}) / \text{Previous avg} \times 100$.
Fare Gap vs Market	$(\text{FreshBus median} - \text{Market median}) / \text{Market median} \times 100$. Negative = FreshBus cheaper.
Occupancy %	$(\text{total_seats} - \text{available_seats}) / \text{total_seats} \times 100$, only when total_seats is valid and > 0 .

5.10 Data Storage

- Primary store: **PostgreSQL** (Docker Compose service for local / staging / production stacks).
- Analytical queries consume a master marketplace view (or equivalent API aggregations), not raw HTML.
- Raw HTML and interim parse artefacts may be retained for audit and reprocessing.
- Historical observations are immutable at the service × snapshot grain.

5.11 Environment Strategy

- **Staging** - Development, testing, validation, and feature verification.
- **Production** - Stable release for end-user access (containerized stack and/or GitHub Pages frontend + API backend as configured).

5.12 Version Control

- Git is used for source code management, version control, and release history.
- Releases move from Staging to Production through controlled deployments.

5.13 Data Refresh Schedule (Summary)

Domain	Schedule
App stores & Google Search Reviews	Daily at 08:00 IST
Redbus marketplace & SRP	Four times daily: 5:00 AM, 11:00 AM, 5:00 PM, 11:00 PM IST
Abhibus marketplace & SRP	Four times daily: 5:00 AM, 11:00 AM, 5:00 PM, 11:00 PM IST (same slots as Redbus)
Full refresh orchestration	Scheduled backend job (aligned to collection windows) plus authorized manual trigger
Dashboard Sync	On demand - reloads stored snapshots only

6. Business Rules

- The dashboard is an internal competitive intelligence platform for FreshBus using publicly available data only.
- Redbus and Abhibus marketplace / SRP data are scraped four times daily at **5:00 AM, 11:00 AM, 5:00 PM, and 11:00 PM IST**.
- Marketplace intelligence is snapshot-based; it is not described as “live data.”
- Sync never triggers scraping.
- SRP lower numerical rank means better visibility; UI copy must state direction (“positions better / worse”).
- Missing data is not displayed as zero; zeros and missing values are different concepts.
- Marketplace date selection cannot go before **01 September 2026**; future dates without completed snapshots are disabled.
- Route-level analysis includes competitors operating three or more services on a particular route where that eligibility rule applies.
- FreshBus is visually identifiable in comparisons without forcing performance colours to green.
- All enhancements are validated in Staging before Production deployment.
- Source code is managed through Git for version control and releases.

7. Time Estimates

Estimates cover delivery of the complete SOW V2.0 scope including Redbus marketplace analytical layer wiring, Abhibus parity, schedule alignment, QA, and deployment.

Life Cycle Stage	Estimated (Hours)	Estimated Completion	Remarks
Requirement Gathering & Analysis	8	07/09/2026	V2.0 scope freeze (marketplace IA, metrics, snapshots)
Documentation & Wire-frame / SOW	12	07/09/2026	This SOW V2.0 + architecture diagrams
Designing (UI / UX analytical system)	24	Based on sign-off	Marketplace, Routes, Operators, SRP Tracker
Front-end implementation	80	Based on sign-off	React dashboard, filters, charts, workbook, drawers
Back-end / scraping / analytical layer	100	Based on sign-off	Collectors, normalize, facts, master view, API
QA Testing & Test cases	40	Based on sign-off	Metric reconciliation, date/snapshot edge cases
UAT & Fixes	24	Based on sign-off	Business stakeholder walkthrough
Deployment	16	Based on sign-off	Staging → Production, Pages / containers
Total	304		

8. Acceptance Criteria (Summary)

- A non-technical business user can understand the Marketplace page without engineering documentation.
- Navigation Marketplace → Route → Operator → Service → Snapshot preserves filter context.
- Every displayed metric has a deterministic definition and can be traced to underlying observations.
- No marketplace data before 01 Sep 2026 appears in the dashboard.
- Four daily snapshot slots are first-class analytical dimensions.
- Sync never initiates scraping; dashboard never scrapes Redbus from the browser.
- SRP charts place #1 at the top; rank improvements are described in positions, not unexplained percentages.

9. Change Request Sign Off

With reference to our discussion for the Peer Competitor Analysis Dashboard, please find annexure for the **SOW #2.0**. Details document enclosed. If you need any additional information, we would be more than happy to provide the same. We are looking forward to your valued order. Kindly sign this SOW contract below and email back.

Udbhav Mishra
SVP - Consumer Strategy & Experience - Freshbus

Ayush Jain
Author - Peer Competitor Analysis Dashboard

10. Version History

Version	Date	Author	Reviewer(s)	Changes
1.0	-	Ayush Jain	Udbhav Mishra	First copy - peer channels, Redbus SRP daily tracking, local Phase 1 framing
2.0	07/09/2026	Ayush Jain	Udbhav Mishra	Comprehensive update: system architecture; Redbus marketplace IA (Marketplace / Routes / Operators / SRP Tracker); 4× daily snapshot model; scrape vs Sync; analytical grain and metrics glossary; Abhibus parity; React/API/Postgres technical approach; acceptance criteria and revised estimates